QUICK-COMMERCE OPERATIONS ANALYSIS

Investigating the cancellation spike at Store GGN_04

SQL-LED DIAGNOSTIC CASE STUDY | MYSQL | STORE OPERATIONS

Total orders	Cancelled orders	Delivered Orders	Overall cancellation rate	Delivery rate
81,333	5,618	75,714	6.91%	93.09%

GGN_04 Total Orders	GGN_04 Cancelled orders	GGN_04 Delivered Orders	GGN_04 cancellation rate	Contribution to total cancellation
10,120	1,215	8,905	12.01%	21.63%

EXECUTIVE TAKEAWAY

GGN_04's cancellation rate rose from 7.53% before October 15 to 23.51% from October 15 onward. At the same time, inventory-related cancellation reasons, recorded stockouts, putaway delays and pick failures all increased. The combined evidence points to an inventory-handling, replenishment and shelf-availability problem as the leading operational hypothesis.

Analysis period: September 1 – October 31, 2026

Prepared as a portfolio case study to demonstrate SQL-led investigation, operational metric interpretation, root-cause hypothesis development and evidence-based recommendations.

1. Executive Summary

The investigation began with 5,618 cancelled orders among 81,333 total orders (6.91%). Store-level analysis identified GGN_04 as the main outlier: its cancellation rate was 12.01%, compared with 6.91% overall, and it contributed 21.63% of all cancellations while handling about 12.45% of orders.

A daily trend showed an abrupt shift on October 15, 2026. The rate rose from 6.74% on October 14 to 23.27% on October 15 and stayed elevated in the following days. Comparing September 1–October 14 (44 days) with October 15–31 (17 days), GGN_04's cancellation rate increased from 7.53% to 23.51%.

LEADING OPERATIONAL HYPOTHESIS

A store-specific inventory handling, replenishment and shelf-availability breakdown at GGN_04 coincided with higher stockouts, putaway delays and pick failures. This is the strongest hypothesis supported by the analysis.

Key indicators at a glance

Metric at GGN_04	Before Oct 15	Oct 15 onward	Change
Cancellation rate (order placed date)	7.53%	23.51%	+15.98 percentage points
Inventory-related cancellation share	14.21%	66.66%	+52.45 percentage points
Stockout rate (inventory snapshots)	4.24%	23.64%	+19.40 percentage points
Complete stockout rate	3.28%	18.78%	+15.50 percentage points
Shelf empty, backroom available	0.96%	4.86%	+3.90 percentage points
Putaway delayed-event rate	7.45%	42.91%	+35.46 percentage points
Backroom processing delay rate	3.12%	29.80%	+26.68 percentage points
Shelf replenishment pending rate	0.24%	13.11%	+12.87 percentage points
Pick failure rate (event-level)	0.39%	6.33%	+5.94 percentage points
Pick delay rate (event-level)	2.73%	6.06%	+3.33 percentage points
Partner unavailability rate	2.92%	2.22%	−0.70 percentage points

Interpretation note: these metrics come from different event populations and timestamp fields. They are complementary signals, not a single causal calculation.

2. Business Problem, Scope & Method

Business problem: investigate where order cancellations are concentrated, identify operational signals associated with the increase, and recommend actions that could improve performance.

Investigation questions

- Which store contributes disproportionately to cancellations?
- When did GGN_04's performance deteriorate, and was the change network-wide?
- Did cancellation reasons shift toward inventory-related issues?
- Did stock availability, putaway processing or picking performance change?
- Does delivery-partner unavailability support a delivery-capacity explanation?

Method and comparison windows

SQL was used to compare overall order outcomes, store-level cancellation rates, daily trends, cancellation-reason shares, inventory snapshot rates, receipts and sales, SKU-level stockouts, putaway outcomes and turnaround, pick-event outcomes and exception reasons, and delivery-partner availability.

The main comparison uses September 1–October 14 (44 days) versus October 15–31 (17 days). Rates and within-period shares are preferred to raw counts because the periods differ in length.

IMPORTANT DISTINCTION

The order cancellation-rate comparison is grouped by order_placed_at; cancellation reason shares use cancellation_timestamp; inventory uses snapshot_date; putaway uses receipt_timestamp; pick performance uses pick_timestamp; and delivery availability uses dispatch_event_timestamp.

3. Overall Performance & Store-Level Outlier

Metric	Result	Interpretation
Total orders	81,333	All orders in the analysis period.
Delivered orders	75,714	Orders recorded as delivered.
Cancelled orders	5,618	Orders recorded as cancelled.
In-progress orders	1	One order remained in progress.
Overall cancellation rate	6.91%	Descriptive baseline; no target benchmark was supplied.
Overall delivery rate	93.09%	Share of orders recorded as delivered.

Store comparison

Store	Orders	Cancelled	Cancellation rate	Share of all cancellations
GGN_04	10,120	1,215	12.01%	21.63%
GGN_08	10,148	652	6.42%	11.61%
GGN_01	10,059	635	6.31%	11.30%
GGN_05	10,349	649	6.27%	11.55%
GGN_03	9,953	624	6.27%	11.11%
GGN_06	10,246	637	6.22%	11.34%
GGN_02	10,267	610	5.94%	10.86%
GGN_07	10,191	596	5.85%	10.61%

WHY GGN_04 WAS PRIORITISED

GGN_04's 12.01% cancellation rate was approximately 1.74 times the 6.91% overall rate. Its 21.63% share of cancellations was disproportionate to its ~12.45% share of orders. Other stores ranged from 5.85% to 6.42%, making GGN_04 a clear outlier.

The overall cancellation rate is a descriptive metric. Without a target or a longer historical benchmark, the analysis does not claim that 6.91% is inherently unacceptable.

4. Timing of the Change & Cancellation Reasons

The daily trend showed GGN_04's cancellation rate increasing from 6.74% on October 14 to 23.27% on October 15, remaining around 19–25% over much of the following period. Other stores had smaller changes overall, although GGN_03 also rose by 1.87 percentage points in the before/after comparison.

Cancellation-reason mix at GGN_04

Cancellation reason at GGN_04	Before Oct 15	Oct 15 onward
ITEM_UNAVAILABLE	6.38%	34.23%
SHELF_REPLENISHMENT_PENDING	7.83%	32.43%
PICKING_DELAY	11.29%	10.51%
DELIVERY_CAPACITY	38.80%	9.46%
PAYMENT_FAILED	8.56%	8.56%
CUSTOMER_CANCELLED	27.14%	4.80%

SHIFT TOWARD INVENTORY-RELATED REASONS

ITEM_UNAVAILABLE plus SHELF_REPLENISHMENT_PENDING increased from 14.21% of recorded cancellation reasons before October 15 to 66.66% from October 15 onward. Before the shift, DELIVERY_CAPACITY was the largest reason category; after the shift, the two inventory-related categories dominated.

These are shares within each period and the cancellation-reason table is based on cancellation timestamps

5. Inventory Availability & Product-Level Signals

Store-level inventory signals

Inventory signal at GGN_04	Before Oct 15	Oct 15 onward	Interpretation
Stockout rate	4.24%	23.64%	Strong deterioration; other stores stayed below 7% post-change.
Shelf empty, backroom available	0.96%	4.86%	More snapshots show stock in backroom but an empty shelf.
Shelf and backroom both empty	3.28%	18.78%	Larger increase, consistent with complete stock depletion.
Average daily receipts	Baseline	−17.48% vs prior period	Receipts declined; investigate supply and timing.
Average daily units sold	Baseline	−18.23% vs prior period	Observed sales also declined; not a direct measure of latent demand.

Both shelf-empty/backroom-available and complete-stockout situations occurred. For example, SKU0038 on October 17 had zero shelf, backroom and system stock; SKU0021 on October 28 had an empty shelf with six units in the backroom. These represent different operational conditions.

SKU-level investigation candidates

Product	Orders containing product	Inventory-related cancelled orders	Rate
SKU0021	320	58	18.13%
SKU0014	340	56	16.47%
SKU0012	320	56	17.50%
SKU0019	276	46	16.67%

Product	Orders containing product	Inventory-related cancelled orders	Rate
SKU0024	268	44	16.42%
SKU0008	129	29	22.48%
SKU0028	127	29	22.83%
SKU0040	148	28	18.92%

The later-period product inventory rates are based on 17 daily snapshots per SKU; Product-level associations do not prove that the product caused an order's cancellation, because orders may contain multiple items and cancellation reasons may not identify a specific SKU.

6. Putaway & Picking Performance

Putaway outcomes

Putaway measure at GGN_04	Before Oct 15	Oct 15 onward
Delayed-event rate	7.45%	42.91%
Backroom processing delay rate	3.12%	29.80%
Shelf replenishment pending rate	0.24%	13.11%
Completed-event rate	92.55%	57.09%
Average overall turnaround	38.67 min	79.24 min
Average delayed-event turnaround	145.76 min	143.93 min
Average completed-event turnaround	30.05 min	30.62 min

The overall putaway turnaround increased from 38.67 to 79.24 minutes (~105%), while average delayed-event turnaround stayed almost unchanged (145.76 to 143.93 minutes) and completed-event turnaround remained stable (30.05 to 30.62 minutes). This suggests the overall increase was driven mainly by delayed events becoming a much larger share of putaway activity, rather than each delayed event taking longer.

Backroom processing delay rate rose from 3.12% to 29.80%, and shelf-replenishment-pending rate rose from 0.24% to 13.11%. GGN_04's delayed-event rate was about 4.7 times the next-highest store's rate in the later period.

Pick-event performance

Pick event metric at GGN_04	Before Oct 15	Oct 15 onward
Total pick events	20,881	7,646
Picked events	20,228	6,699
Failed events	82	484
Delayed events	571	463

Pick event metric at GGN_04	Before Oct 15	Oct 15 onward
Picked rate	96.87%	87.61%
Failure rate	0.39%	6.33%
Delay rate	2.73%	6.06%

The pick failure rate increased from 0.39% to 6.33%, and the pick delay rate from 2.73% to 6.06%. The total failed events increased from 82 to 484 even though the later period was shorter; rates remain the better period comparison. These are pick-event metrics, not order-level failure rates.

Recorded pick exception reasons

Pick exception	Before Oct 15	Oct 15 onward
Failed — ITEM_UNAVAILABLE	36 (43.90% of failed)	243 (50.21% of failed)
Failed — SHELF_REPLENISHMENT_PENDING	46 (56.10% of failed)	241 (49.79% of failed)
Delayed — ITEM_HARD_TO_LOCATE	209 (36.60% of delayed)	161 (34.77% of delayed)
Delayed — PICKER_BACKLOG	192 (33.63% of delayed)	137 (29.59% of delayed)
Delayed — SHELF_REPLENISHMENT_PENDING	170 (29.77% of delayed)	165 (35.64% of delayed)

PICKING INTERPRETATION

After October 15, failed picks were split between ITEM_UNAVAILABLE (50.21%) and SHELF_REPLENISHMENT_PENDING (49.79%). Shelf replenishment also represented a larger share of delayed picks (35.64%, up from 29.77%). This aligns with the inventory signals but does not independently prove causation.

7. Delivery Availability & Root-Cause Assessment

Delivery-partner availability

Delivery measure	Before Oct 15	Oct 15 onward
Delivery events	7,287	2,833
Assigned events	6,766	2,170
Partner unavailable events	213	63
Partner unavailability rate	2.92%	2.22%

Recorded partner unavailability decreased from 2.92% to 2.22%. This makes partner unavailability a less convincing primary explanation for the cancellation spike. The metric does not rule out other delivery issues, including delayed assignment or late deliveries.

Evidence synthesis

- Cancellation rate at GGN_04 increased from 7.53% to 23.51%.
- Inventory-related cancellation share increased from 14.21% to 66.66%.
- Stockout rate increased from 4.24% to 23.64%; complete stockouts increased from 3.28% to 18.78%.
- Putaway delayed-event rate increased from 7.45% to 42.91%, with backroom processing and shelf replenishment delay rates rising sharply.
- Pick failure rate increased from 0.39% to 6.33%; recorded failures were inventory-related.
- Partner unavailability fell from 2.92% to 2.22%, so it does not match the direction of the cancellation spike.

ROOT-CAUSE ASSESSMENT

The strongest working hypothesis is a store-specific inventory handling and replenishment problem at GGN_04, potentially reducing shelf availability and contributing to pick failures and inventory-related cancellations.

8. Recommendations & Validation Plan

1 — Immediate | Fix shelf replenishment delays

Check why received products are delayed in moving from the backroom to store shelves at GGN_04.

Monitor: ***Delayed putaway rate; receipt-to-shelf availability time; pending replenishment by shift.***

2 — Immediate | Prioritise affected SKUs

Investigate SKUs such as SKU0021, SKU0038 and SKU0040 to identify stock shortages and replenishment issues.

Monitor: ***Stockout rate and inventory-related cancellation rate by SKU and day.***

3 — Immediate | Review pick failures

Find out why pickers cannot find products—whether stock is unavailable or products are still in the backroom.

Monitor: ***Pick failure rate by exception reason; shelf replenishment pending rate.***

4 — Introduce preventive stock checks

Conduct regular shelf and backroom stock checks, especially before peak hours, to identify shortages and replenish products before orders are placed.

Monitor: ***Stock availability and stockout rate by shift.***

5 — Add early-warning monitoring

Monitor cancellations and inventory signals daily by store and reason, with alerts for sustained deviations from each store's baseline.

Monitor: ***Cancellation rate; inventory-related cancellation share; stockout and putaway delay rates.***

6 — Establish clear replenishment ownership

Assign staff responsibility for monitoring pending replenishment, resolving backroom processing delays and ensuring products reach shelves within a defined time.

Monitor: ***Pending replenishment, resolution time and delayed putaway rate.***

Proposed operational target (not a company standard)

RECOVERY DIRECTION

Use GGN_04's pre-change cancellation rate of 7.53% as a reference point for a recovery objective, while validating feasibility with operations. Track cancellation, stockout, putaway delay and pick failure rates together so that a reduction in cancellations is not achieved at the expense of another metric.

9. Limitations & Conclusion

Limitations

Different timestamps and event populations

Orders, cancellation events, inventory snapshots, putaway, pick and delivery records use different timestamps and granularities. They are complementary signals, not a fully aligned causal timeline.

Unequal comparison windows

The before period is 44 days and the after period is 17 days. Rates and within-period shares are more appropriate than raw counts.

Product attribution

Orders can contain multiple products; a product appearing in a cancelled order is not proof that it caused the cancellation.

Demand is not directly observed

Observed sales can be suppressed by stockouts. Lower sales do not necessarily mean underlying demand fell.

Inventory accuracy check is one-sided

A zero rate for system stock exceeding physical stock does not detect the opposite discrepancy or prove all inventory records are accurate.

Conclusion

FINAL CONCLUSION

GGN_04 experienced a sharp increase in order cancellations from October 15, 2026, compared with the other stores. The increase coincided with higher stockouts, shelf replenishment delays and pick failures, indicating that inventory availability and replenishment processes are the key areas requiring improvement.

To prevent similar incidents, the store should improve the movement of products from receiving to shelves, prioritise frequently unavailable products and conduct regular stock checks. These measures should be monitored through daily operational metrics to assess their effectiveness and reduce future cancellations.

This report is based on the recorded findings from the SQL investigation.